UGST4085 — Numerical Methods in Calculus and Algebra
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Finite difference schemes for two-variable functions and their sociological application

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https://github.com/cseresznyeszeteny/UGST4085_FDS

1 Finite Difference Schemes in One and Two dimensions

1.1 Introduction

Many partial differential equations that model physical phenomena cannot be solved purely analytically. Examples include heat conduction, electrostatics, fluid flow, option pricing, but also social phenomena: how information spreads in social networks. Finite difference schemes approximate derivatives by differences of function values on a discrete grid. This converts a continuous partial differential equation problem into a finite–dimensional algebraic system, more typically a sparse linear system, which can be solved efficiently on a computer.

The advantages of using finite differential schemes to solve these problems are that they are conceptually simple and can be based directly on Taylor expansions, that their implementation is easy on structured grids in one, two, or three dimensions, because they naturally produce sparse matrices with local stencils. In this section, we review finite difference formulas in one and two dimensions, derive the classical 5-point stencil for the Laplacian equation, and introduce the basic classification of second-order partial differential equations.

1.2 Finite Difference Formulas in One Dimension

Consider a smooth function $u(x)$ on an interval $[a, b]$ and a uniform grid $x_i = a+ih$, for $i = 0, 1, \ldots, N$, with $h = \frac{b-a}{N}$. To approximate derivatives, we use forward, backward, and central finite difference formulas for the first derivative, and the central second-difference operator for the second derivative.

A one-dimensional finite difference scheme discretizes the domain, replaces derivatives with these difference formulas, incorporates boundary conditions, and produces a linear system $Au = B$ which is able to approximate the differential equation.

1.3 Finite Differences in Two Dimensions and the 5-point Stencil

Let $u(x, y)$ be a smooth function defined on a rectangular (often square) domain, for example $\Omega = (0, 1) \times (0, 1)$. We denote $u_{i,j} := u(x_i, y_j)$ and introduce a uniform grid

$$x_i = i \cdot h, \quad i = 0, \ldots, N+1, \qquad y_j = j \cdot h, \quad j = 0, \ldots, N+1, \qquad \text{where } h = \frac{1}{N+1}.$$

Approximations of u_{xx} and u_{yy}

At an interior grid point (x_i, y_j) we approximate the second derivatives independently in each direction:

$$u_{xx}(x_i, y_j) \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2},$$
$$u_{yy}(x_i, y_j) \approx \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2}.$$

Both approximations are second–order accurate:

$$u_{xx}(x_i, y_j) = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \mathcal{O}(h^2), \qquad u_{yy}(x_i, y_j) = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} + \mathcal{O}(h^2).$$

Derivation of the 5-point stencil for the Laplacian equation

The Laplace operator in two dimensions is

$$\Delta u = u_{xx} + u_{yy}.$$

Combining the finite difference formulas above, we obtain

$$\begin{aligned}\Delta u(x_i, y_j) &\approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} \\ &= \frac{u_{i+1,j} + u_{i-1,j} - 4u_{i,j} + u_{i,j+1} + u_{i,j-1}}{h^2}.\end{aligned}$$

Thus a finite difference approximation of the negative Laplacian at (x_i, y_j) is

$$-\Delta u(x_i, y_j) \approx \frac{-u_{i+1,j} - u_{i-1,j} + 4u_{i,j} - u_{i,j+1} - u_{i,j-1}}{h^2}.$$

This is the classical 5-point stencil: each interior point interacts only with its four nearest neighbors (left, right, up, down) and itself.

$$\begin{array}{ccccc} & & u_{i,j+1} & & \\ & & | & & \\ u_{i-1,j} & - & u_{i,j} & - & u_{i+1,j} \\ & & | & & \\ & & u_{i,j-1} & & \end{array}$$

Example: Poisson equation on the unit square

Consider the Poisson problem

$$-\Delta u = 1 \quad \text{on } \Omega = (0, 1)^2, \qquad \text{and } u = 0 \quad \text{on } \partial\Omega.$$

At an interior grid point (x_i, y_j), $i = 1, \ldots, N$, and $j = 1, \ldots, N$ the finite difference scheme is

$$\frac{-u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1} + 4u_{i,j}}{h^2} = 1.$$

Boundary values are prescribed by the Dirichlet condition $u = 0$ on $\partial\Omega$. Collecting all interior unknowns $u_{i,j}$ into a vector leads to a sparse linear system $A\mathbf{u} = \mathbf{b}$ with a block tridiagonal structure.

1.4 Classification of Second–Order Partial Differential Equations

A general linear second–order partial differential equation in two variables can be written as

$$A(x,y)u_{xx} + 2B(x,y)u_{xy} + C(x,y)u_{yy} + D(x,y)u_x + E(x,y)u_y + F(x,y)u = G(x,y),$$

where A, B, C, D, E, F, G are given coefficient functions. The classification is based on the *discriminant*

$$\Delta = B^2 - AC.$$

At a given point (x, y) we distinguish three cases:

- If $\Delta < 0$, it is an elliptic equation, with typical examples of the Laplace or Poisson equations. Then, $-\Delta u = f$.
- If $\Delta = 0$, it is a parabolic equation, with a typical example of the heat equation. Then $u_t - \alpha\Delta u = 0$.
- If $\Delta > 0$, it is a hyperbolic equation, with a typical example of the wave equation. Then $u_{tt} - c^2\Delta u = 0$.

2 Discretization of the Two-Dimensional Domain and Matrix Formulation

2.1 The Argument for Discretization

To numerically solve partial differential equations such as the Poisson or Laplace equation,

$$-\Delta u = f \qquad \text{on } \Omega \subset \mathbb{R}^2,$$

we must replace the continuous domain Ω by a finite set of grid points. See the discretization we considered for the uniform grind mentioned in section 1.3 for example. This allows us to approximate derivatives using finite differences, convert the partial differential equations into a system of algebraic equations, to impose boundary conditions directly on grid nodes, and to solve the resulting sparse linear system using iterative or direct methods. The discretization is therefore the bridge between the continuous mathematical problem and the computable numerical approximation.

2.2 Indexing and Vectorization

To write the discrete problem as a linear system, we map the two-dimensional grid of unknowns to a one-dimensional vector using row-major ordering:

$$u_{1,1},\ u_{2,1},\ \ldots,\ u_{N,1},\ u_{1,2},\ \ldots,\ u_{N,N}.$$

The resulting vector is $u \in \mathbb{R}^{N^2}$. Since the stencil produces one algebraic equation per interior point, we obtain N^2 equations for the N^2 unknowns $u_{n,m}$. Linear algebra solvers operate on vectors, so the grid must be flattened, allowing the discrete partial differential equation to be written compactly as $Au = b$, where A encodes the 5-point stencil. A grid point (n, m) with $n, m = 1, \ldots, N$ is mapped to a single vector index by

$$\boxed{i = n + (m-1)N}.$$

Thus the first N entries correspond to the first row of the grid, the next N to the second row, and so on. The inverse mapping $n = 1 + (i-1) \bmod N$ and $m = 1 + \lfloor (i-1)/N \rfloor$ restores (n, m) from i.

Consequence for the Linear System

Because $u \in \mathbb{R}^{N^2}$, the discrete PDE leads to a matrix $A \in \mathbb{R}^{N^2 \times N^2}$, with one row per interior stencil equation. Under the mapping $i = n + (m-1)N$, the neighbors $u_{n\pm1,m}$ and $u_{n,m\pm1}$ become $u_{i\pm1}$ and $u_{i\pm N}$. Each row therefore contains at most five nonzero entries, giving A its characteristic sparse block-tridiagonal structure.

Role of Boundary Condition.

The finite difference stencil applies only to interior points. Boundary nodes do not satisfy stencil equations; instead, their values are fixed by the boundary conditions. Without boundary conditions the system would be underdetermined, because the PDE alone does not specify the solution on the boundary. For Dirichlet boundaries, where u is prescribed on $\partial\Omega$, the corresponding rows of A simply enforce

$$u_i = g_i,$$

which is implemented by placing a 1 on the diagonal of that row and zeros elsewhere, and setting the corresponding entry of b equal to g_i. For Neumann boundaries, where the normal derivative is given, one modifies the stencil or the right–hand side to incorporate the derivative condition.

Example: Why Boundary Conditions Are Necessary and Recovering the Numerical Solution.

To see concretely why boundary conditions are required, consider the finite–difference discretization of the equation

$$\frac{f_{n-1,m} - 2f_{n,m} + f_{n+1,m}}{h^2} + \frac{f_{n,m-1} - 2f_{n,m} + f_{n,m+1}}{h^2} = \left(-\frac{2}{\alpha} - \frac{2}{\beta} + \frac{4x_n^2}{\alpha^2} + \frac{4y_m^2}{\beta^2}\right) f_{n,m},$$

which must hold for every *interior* grid point (n, m). For simplicity let us take a 3×3 set of interior points ($N = 3$), corresponding to the indices

$$(n, m) \in \{1, 2, 3\} \times \{1, 2, 3\}.$$

Step 1: Flattening the Grid. Using the index mapping $i = n + (m-1)N$, the 3×3 interior grid is mapped to the vector

$$f = (f_{1,1},\ f_{2,1},\ f_{3,1},\ f_{1,2},\ f_{2,2},\ f_{3,2},\ f_{1,3},\ f_{2,3},\ f_{3,3})^{\mathsf{T}} \in \mathbb{R}^9.$$

Step 2: Building the Matrix Without Boundary Conditions. Applying the finite difference stencil to each interior point produces a system of nine equations. For example, the equation at the point $(n, m) = (2, 2)$ (i.e. at the center of the grid) becomes

$$\frac{f_{1,2} - 2f_{2,2} + f_{3,2}}{h^2} + \frac{f_{2,1} - 2f_{2,2} + f_{2,3}}{h^2} = \left(-\tfrac{2}{\alpha} - \tfrac{2}{\beta} + \tfrac{4x_2^2}{\alpha^2} + \tfrac{4y_2^2}{\beta^2}\right) f_{2,2}.$$

Using the mapping $i = n + (m-1)N$, this equation involves the vector entries $f_5,\ f_2,\ f_8,\ f_4,\ f_6$, corresponding to $(2, 2)$, $(1, 2)$, $(3, 2)$, $(2, 1)$, $(2, 3)$. Thus the fifth row of the matrix A has the form

$$A_{5,2} = A_{5,4} = A_{5,6} = A_{5,8} = \frac{1}{h^2}, \qquad A_{5,5} = -\frac{4}{h^2} - \frac{2}{\alpha} - \frac{2}{\beta} + \frac{4x_2^2}{\alpha^2} + \frac{4y_2^2}{\beta^2},$$

and zeros elsewhere. However, for points on the *interior boundary* of the unknown region, such as $(n, m) = (1, 1)$, the stencil formally requires the values $f_{0,1}, f_{2,1}, f_{1,0}, f_{1,2}$. The values $f_{0,1}$ and $f_{1,0}$ lie *outside* the interior grid and are not unknowns. Hence the corresponding rows of A cannot be completed.

The resulting matrix therefore has the schematic form

$$A = \begin{pmatrix} * & & & \cdots \\ & * & & \\ \vdots & & \ddots & \\ & & & * \end{pmatrix},$$

where some entries marked $*$ cannot yet be filled because the stencil depends on values that are not part of the unknown vector f. Thus, the system $Af = b$ is incomplete and underdetermined.

Step 3: Imposing Boundary Conditions Completes the System. Suppose we impose Dirichlet boundary conditions, for example $f = g(x, y)$ on $\partial\Omega$. Then values such as $f_{0,1}$ and $f_{1,0}$ are known numbers and can be moved to the right–hand side of the interior equations. For the corner equation $(n, m) = (1, 1)$ we obtain, after substituting boundary values,

$$\frac{g_{0,1} - 2f_{1,1} + f_{2,1}}{h^2} + \frac{g_{1,0} - 2f_{1,1} + f_{1,2}}{h^2} = \left(-\frac{2}{\alpha} - \frac{2}{\beta} + \frac{4x_1^2}{\alpha^2} + \frac{4y_1^2}{\beta^2}\right) f_{1,1}.$$

The previously "missing" entries in the matrix A are now fully determined, because the terms $g_{0,1}$ and $g_{1,0}$ appear in the corresponding positions of the vector b. This completes the algebraic system, making it square and solvable.

Conclusion. This example illustrates that the stencil at interior points requires values at neighboring nodes, some of which lie on the boundary. Without boundary conditions these values are unknown, and the linear system cannot be completed. Once the boundary values are specified (Dirichlet) or expressed through derivatives (Neumann), the missing terms are supplied and the full system $Af = b$ becomes well posed. Once the linear system $Au = b$ is solved, the components of the vector u provide approximations to the P.

2.3 Stability, Consistency, Convergence, and Iterative Solvers

A finite difference scheme is consistent with a partial differential equation if its local truncation error tends to zero as $h \to 0$. For the two-dimensional Laplacian, the central second–difference formulas

$$u_{xx}(x_i, y_j) = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \mathcal{O}(h^2), \qquad u_{yy}(x_i, y_j) = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} + \mathcal{O}(h^2)$$

show that the 5-point stencil is second–order accurate. Stability ensures that numerical errors do not grow uncontrollably. For elliptic problems such as $-\Delta u = f$, the discrete Laplacian produces a symmetric positive definite matrix A, so the linear system $Au = b$ is stable and its inverse is bounded. By the Lax Equivalence Theorem, for linear well–posed problems,

$$\text{Consistency} + \text{Stability} \;\Rightarrow\; \text{Convergence}.$$

Since the scheme is second–order consistent and A is SPD, the numerical solution u_h satisfies

$$\|u - u_h\| = \mathcal{O}(h^2),$$

and the global error reflects the $\mathcal{O}(h^2)$ local truncation error where global error is the difference between the numerical solution and the exact solution of the partial differential equation. The local error measures how well a single difference formula approximates a derivative.

Finally, the conditioning of the discrete Laplacian scales like $\kappa(A) = \mathcal{O}(h^{-2})$, meaning finer grids increase ill-conditioning, but learned iterative solvers remain effective due to the symmetric positive definite structure of A.

3 Outlook on Sociological Applications: Rumor Spreading

3.1 Introduction and Motivation

Rumors play an important role in social dynamics. They influence behavior, shape public opinion, and contribute to group cohesion or fragmentation. The spread of rumors is driven by local interactions: individuals primarily share information with those who are socially or spatially close to them. This mechanism is strikingly similar to diffusion processes in physics, where heat or chemicals spread through a medium via local exchanges.

Because of this structural similarity, partial differential equations provide a natural mathematical framework for modeling the spread of rumors. They describe how a quantity—such as awareness or belief—changes in space and time. In this case, the quantity of interest is the proportion of the population aware of the rumor at each location, which evolves both through local sharing (reaction) and through spatial proximity (diffusion). This allows parameters like communication intensity or social connectivity to be captured in the model.

To solve such models numerically, we use finite difference schemes. These are classic numerical methods that approximate the derivatives in a partial differential equation using discrete differences on a grid. For example, second-order spatial derivatives like those in the Laplacian are replaced by combinations of neighboring grid values—commonly via the five-point stencil in two dimensions. Time derivatives can be approximated using explicit or implicit schemes, such as the forward Euler method.

Finite dIfference schemes are particularly attractive because they are simple to implement, computationally efficient, and conceptually intuitive. They also allow us to simulate the behavior of a model over time and visualize how information spreads spatially across a population.

In this project, we extended finite difference schemes to two-variable functions and apply them to a sociological model describing how a rumor diffuses throughout a community. Our focus is on interpreting the mathematical results in a sociological context, using numerical simulations to illustrate how local interaction rules can lead to large-scale patterns of awareness.

3.2 The Rumor Spreading Model Setup

The spread of rumor awareness using a reaction–diffusion equation:

$$\frac{\partial u}{\partial t} = D\nabla^2 u + u(i,j)$$

where:

- $u(i,j)$ represents the spreading function of the rumor, which gives the extent of aweraness of the rumor in each gridpoint;
- $D > 0$ is the diffusion coefficient, governing spatial spread due to local social interactions;
- $\nabla^2 u$ is the Laplacian, approximating how rumor awareness flows between neighboring regions.

The diffusion term $D\nabla^2 u$ models interpersonal interactions across space: individuals hear the rumor from nearby contacts. The reaction term $u(i,j)$ mimics a logistic growth function: spread slowly accelerates when few people know the rumor, but then spread explodes until awareness saturates and fewer new individuals can be informed. This logistic structure aligns with classical sociological rumor models, though our continuous partial differential equation approach captures spatial dynamics not present in those discrete models.

3.3 Model Implementation

For the simulation, we define parameters as the following: the space of the simulation will be an $n \times n$ grid, D will be the diffusion coefficient, ∂t will be the time step, ε is a an adjustment to avoid division

by zero, u_m is the maximum spreading, α and β are exponents to ensure nonlinearity in the simulation, $\mathcal{N}_{i,j}$ is the average of neigbors, the rumor growth function is an empirically designed $u(i,j)$.

$$u(i,j) = (\mathcal{N}_{i,j} + \varepsilon)^{\alpha} \cdot \max|u_m - u(i,j), 0|^{\beta} - u(i,j)\left(1 - \frac{\mathcal{N}_{i,j}}{u_m + \varepsilon}\right), \text{ where}$$

$$n = 100, D = 0.3, \partial t = 0.75, \varepsilon = 10^{-12}, u_m = 1.0, \alpha = 6, \beta = 3.$$

Therefore, the function describing the simulation of rumor spreading is defined similarly as above:

$$\frac{\partial u}{\partial t} = D \cdot \nabla^2 u + u(i,j)$$

The reaction term, or rumor spread function is designed in such way, that it takes into account the values of its neighbors, and it increases more with higher function values. The Python implementation of the rumor spread model can be found in the GitHub repository referenced below the title.

3.4 Model Dynamics Over Time

The simulation of the rumor spread model reveals three distinct phases that reflect the structure of the reaction–diffusion equation and the chosen parameters. In each phase, the dynamics are governed by the balance between the diffusion term $D\nabla^2 u$, which spreads awareness across neighboring grid points, and the nonlinear reaction term $u(i,j)$, which depends on the local neighborhood average $N_{i,j}$, the exponents α and β, and the maximum awareness level u_m.

Early Phase. The rumor is introduced at one or a few seed locations, so initially most grid points have awareness values $u(i,j)$ close to zero, and their neighborhood averages $N_{i,j}$ are also very small. In this regime, the empirically designed reaction term contributes little net growth, since both $u(i,j)$ and $N_{i,j}$ are far from the range where the nonlinear factors become large. The evolution of awareness is therefore slow and mainly driven by diffusion, which creates a smooth, approximately circular region of elevated awareness around the initial seeds.

Intermediate Phase (Rapid Spread). As time progresses, awareness around the seed locations increases, and both the pointwise values $u(i,j)$ and the neighborhood averages $N_{i,j}$ enter a range where the nonlinear reaction term becomes large and positive. In areas where many neighbors are already aware, the contribution of the reaction term is amplified, while the dependence on the gap to u_m ensures that growth is strongest where awareness is still noticeably below the maximum level. Together with diffusion, this produces a rapidly expanding wavefront and coherent clusters of high awareness that grow and spread across the grid.

Late Phase (Saturation). When most grid points and their neighborhoods have awareness values close to the maximum level u_m, the structure of the reaction term causes growth to slow down. The factor that depends on the difference between $u(i,j)$ and u_m becomes small as this gap closes, and the damping part of the reaction further reduces net change in regions where both the point and its neighbors are highly aware. The dynamics are then dominated by diffusion, which mainly smooths any remaining spatial variations. In the simulations, the system approaches a nearly steady state where the absolute value of $\partial u/\partial t$ is small and $u(i,j)$ is close to u_m across the grid, although minor fluctuations may persist due to the nonlinear interaction structure.

3.5 Interpretation from a Sociological Perspective

The parameters in the model admit clear sociological interpretations when seen as stylized descriptions of local interactions on a spatial or social grid. The diffusion coefficient D measures how strongly

awareness spreads purely through proximity. A larger D corresponds to a more interconnected or mobile community where information travels quickly between neighboring locations, while a smaller D represents more localized, slower communication. The exponents α and β, together with the neighborhood average $N_{i,j}$ and the maximum level u_m, encode how strongly individuals respond to the awareness of their neighbors, and how sharply growth accelerates or slows as local awareness increases.

A larger α describes a more "herd-like" response, where moderate increases in neighbor awareness can trigger disproportionately strong local growth. The saturation behavior reflects the idea that once both an individual and their neighbors are nearly fully aware, there is little room for further increase in awareness or spreading activity, so the local change becomes small. The simulated spreading pattern—awareness expanding symmetrically from a single seed in a homogeneous grid—illustrates how a rumor starting in a structurally central or well-connected region can reach a large portion of the population relatively quickly. The near-saturation seen in the simulations, where $u(i, j)$ approaches $u_m = 1$ at nearly every location, indicates a theoretical scenario in which almost all individuals are aware of the rumor. Although this level of complete awareness is uncommon in actual societies, the model illustrates how straightforward local rules and network connections can produce extensive knowledge. Differences in the speed and extent of spread under alternative values for D, α, β, and u_m can be interpreted as stylized representations of unequal access to information, varying communication norms, or differing levels of social cohesion across communities.

3.6 Numerical Behaviour and Interpretation

The observed numerical patterns—such as approximately circular expansion of awareness from the initial seed, the formation of a pronounced wavefront during the rapid-spread phase, and the gradual leveling-off toward a high-awareness state—are direct consequences of the specific reaction–diffusion structure and the finite difference discretization used in the simulations. The symmetry of the spreading front follows from the use of a regular grid, a constant diffusion coefficient D, and spatially homogeneous reaction parameters applied at every grid point, so there is no built-in directional bias or heterogeneity in the numerical setup.

At the same time, these outcomes must be interpreted as idealized rather than literal predictions. The near-uniform high values of $u(i, j)$ reached in the simulations correspond to the system evolving toward a state where most points are close to the maximum awareness level $u_m = 1$. This is consistent with the saturation behavior of the nonlinear reaction term, but relies on the homogeneous and deterministic nature of the model. In actual social systems, varied network structures, different individual actions, external factors, and chance are likely to disrupt the symmetries observed here and hinder complete saturation. Consequently, the numerical outcomes ought to be regarded mainly as a clear exemplification of how the selected parameters D, $u(i, j)$, $N_{i,j}$, α, β, and u_m work together to generate intricate patterns, instead of being seen as forecasts for a particular community.

3.7 Limitations

While the finite difference model gives a useful approximation of how a rumor spreads through local interactions, it also simplifies many aspects of real social behavior. However we are aware that anything further at the moment would be either out of the scope of course or too complex to model with the means feasible. This section aims to mention possible concerns raised with respect to the model built up.

The model assumes a regular grid structure of people arranged on a perfect two-dimensional grid, interacting only with their four immediate neighbors. Real social interactions are much more complex and irregular.

Every individual is treated the same in the simulation, and this means that same likelihood to spread the rumor, the same is the reaction to receiving it. In reality, people vary in how much they trust, share, or even believe a rumor. Even if someone becomes aware of the rumor, in the model,

there is the underlying assumption that they stay aware forever. There's no mechanism for forgetting, changing belief, or randomly stopping the spread.

The model starts with small rumor source. In reality, rumors can start in multiple places, evolve over time, or be challenged by other information. The model does not capture this complexity.

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